1D CNN — Architecture Variant Comparison

Generated 2026-05-14 22:53:54

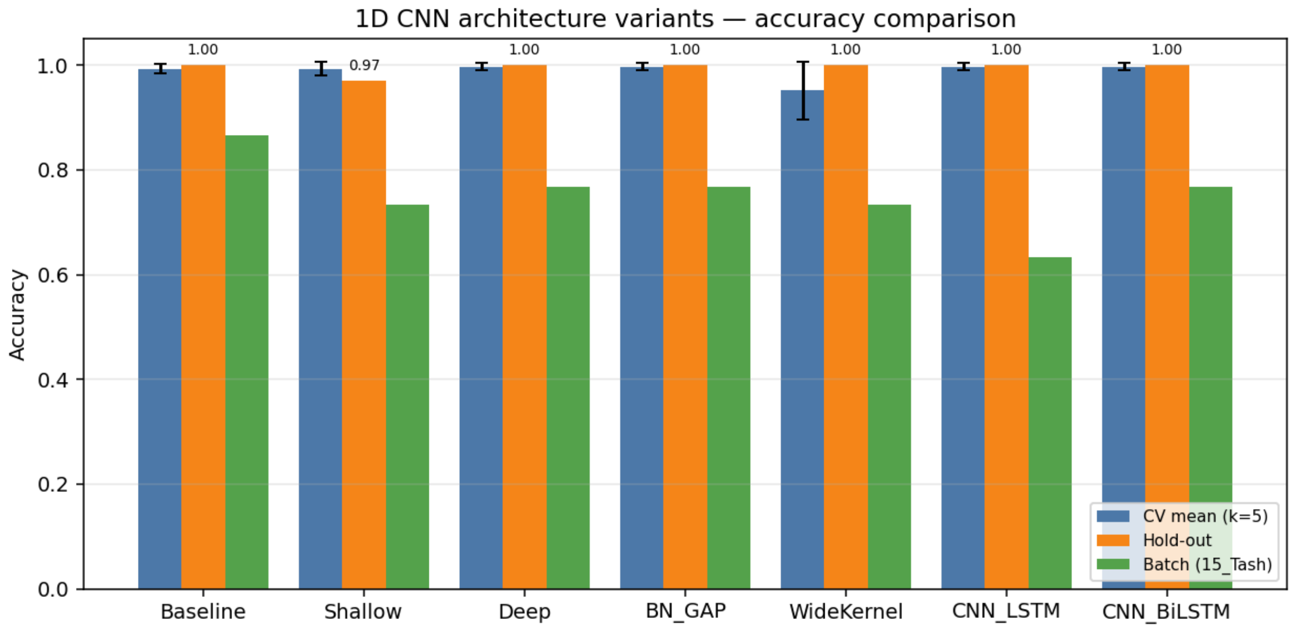
Data & configuration

Training path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic
Training path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic
Training path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic
Training path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic
Training path 5	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic
Training path 6	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic
Training path 7	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic
Training path 8	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic
Training path 9	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic
Training path 10	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic
Training path 11	/home/jestin/ThesisRepo/ML/NewTestData/18_Bridgette/Dynamic
Batch-test root	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Dynamic
Total trials	325 (train pool 292, hold-out 33)
Classes	6: Double_Lower, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	176 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 10.0 Hz, order 4
Normalisation	minmax
CV folds · shuffle	5 · True
Augmentation	on · copies=2 · amplitude_scale, baseline_offset
Epochs · batch size	60 · 16

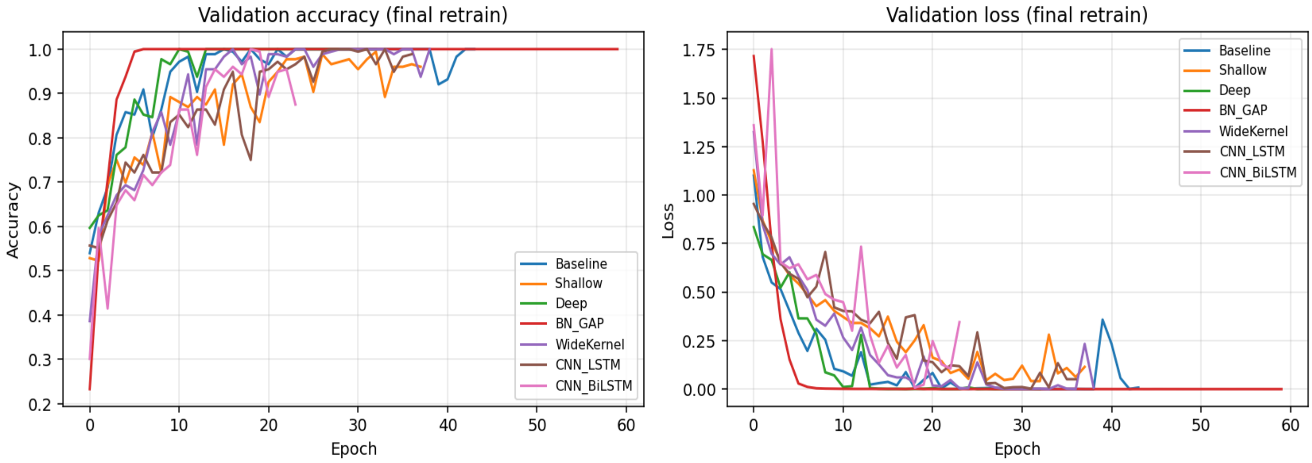
Variant comparison (summary)

Variant	Params	CV mean (k=5)	CV std	Hold-out acc	Hold-out loss	Batch acc	Batch n
Baseline	113,190	0.9932	0.0084	1.0000	0.0001	0.8667	26/30
Shallow	72,454	0.9932	0.0136	0.9697	0.0501	0.7333	22/30
Deep	203,878	0.9966	0.0068	1.0000	0.0001	0.7667	23/30
BN_GAP	57,190	0.9966	0.0068	1.0000	0.0000	0.7667	23/30
WideKernel	131,622	0.9518	0.0550	1.0000	0.0001	0.7333	22/30
CNN_LSTM	68,390	0.9966	0.0068	1.0000	0.0017	0.6333	19/30
CNN_BiLSTM	105,510	0.9966	0.0068	1.0000	0.0036	0.7667	23/30

Best on hold-out: **Baseline** · Best on CV: **Deep**



Training curves (final retrain on full training pool)



Validation accuracy (final retrain)
Accuracy
Epoch
Baseline
Shallow
Deep
BN_GAP
WideKernel
CNN_LSTM
CNN_BiLSTM
Validation loss (final retrain)
Loss
Epoch

Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	113,190
CV mean ± std	0.9932 ± 0.0084
Hold-out test acc / loss	1.0000 · 0.0001
Batch-test acc	0.8667 (26/30)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	699	59	1.0000	1.0000	1.0000
2	699	59	0.9831	0.9831	0.9831
3	702	58	1.0000	1.0000	0.9828
4	702	58	0.9828	0.9828	0.9828
5	702	58	1.0000	1.0000	1.0000

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	1.000	1.000	1.000	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	*1.000*	*1.000*	*1.000*	*33*
weighted avg	*1.000*	*1.000*	*1.000*	*33*

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	4	5	0.800
Drum_Roll	2	5	0.400
Overall	**26**	**30**	**0.867**

Variant: Shallow

Single Conv block: Conv1D(32,k=5)→Pool→Flatten→Dense(32)→Dropout(0.3)→Softmax

Trainable params	72,454
CV mean ± std	0.9932 ± 0.0136
Hold-out test acc / loss	0.9697 · 0.0501
Batch-test acc	0.7333 (22/30)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	699	59	1.0000	1.0000	1.0000
2	699	59	0.9661	0.9831	0.9661
3	702	58	1.0000	1.0000	0.9655
4	702	58	1.0000	1.0000	0.9828
5	702	58	1.0000	1.0000	1.0000

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	0.833	1.000	0.909	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	1.000	0.800	0.889	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	*0.972*	*0.967*	*0.966*	*33*
weighted avg	*0.975*	*0.970*	*0.969*	*33*

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	2	5	0.400
Drum_Roll	0	5	0.000
Overall	**22**	**30**	**0.733**

Variant: Deep

Three Conv blocks: Conv1D(32,4)→Conv1D(64,4)→Conv1D(128,3)→Flatten→Dense(128)→Dropout(0.4)→Softmax

Trainable params	203,878
CV mean ± std	0.9966 ± 0.0068
Hold-out test acc / loss	1.0000 · 0.0001
Batch-test acc	0.7667 (23/30)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	699	59	1.0000	1.0000	0.9492
2	699	59	0.9831	0.9831	0.9831
3	702	58	1.0000	1.0000	1.0000
4	702	58	1.0000	1.0000	0.9828
5	702	58	1.0000	1.0000	0.9828

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	1.000	1.000	1.000	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	*1.000*	*1.000*	*1.000*	*33*
weighted avg	*1.000*	*1.000*	*1.000*	*33*

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	4	5	0.800
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	2	5	0.400
Drum_Roll	2	5	0.400
Overall	**23**	**30**	**0.767**

Variant: BN_GAP

Conv+BN+ReLU ×3 with GlobalAveragePooling1D head — fewer params, less overfit

Trainable params	57,190
CV mean ± std	0.9966 ± 0.0068
Hold-out test acc / loss	1.0000 · 0.0000
Batch-test acc	0.7667 (23/30)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	699	59	1.0000	1.0000	1.0000
2	699	59	0.9831	0.9831	0.9831
3	702	58	1.0000	1.0000	0.9483
4	702	58	1.0000	1.0000	1.0000
5	702	58	1.0000	1.0000	1.0000

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	1.000	1.000	1.000	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	*1.000*	*1.000*	*1.000*	*33*
weighted avg	*1.000*	*1.000*	*1.000*	*33*

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	4	5	0.800
Double_Pistol_Recoil	5	5	1.000
Double_Raise	4	5	0.800
Double_Wiggle	5	5	1.000
Double_cmere	4	5	0.800
Drum_Roll	1	5	0.200
Overall	**23**	**30**	**0.767**

Variant: WideKernel

Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	131,622
CV mean ± std	0.9518 ± 0.0550
Hold-out test acc / loss	1.0000 · 0.0001
Batch-test acc	0.7333 (22/30)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	699	59	1.0000	1.0000	1.0000
2	699	59	0.9831	0.9831	0.9661
3	702	58	0.8621	0.8966	0.8966
4	702	58	1.0000	1.0000	0.9828
5	702	58	0.9138	0.9138	0.8276

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	1.000	1.000	1.000	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	*1.000*	*1.000*	*1.000*	*33*
weighted avg	*1.000*	*1.000*	*1.000*	*33*

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	4	5	0.800
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	2	5	0.400
Drum_Roll	1	5	0.200
Overall	**22**	**30**	**0.733**

Variant: CNN_LSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→LSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	68,390
CV mean ± std	0.9966 ± 0.0068
Hold-out test acc / loss	1.0000 · 0.0017
Batch-test acc	0.6333 (19/30)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	699	59	1.0000	1.0000	0.8475
2	699	59	0.9831	0.9831	0.9831
3	702	58	1.0000	1.0000	0.9655
4	702	58	1.0000	1.0000	0.8966
5	702	58	1.0000	1.0000	0.9828

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	1.000	1.000	1.000	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	*1.000*	*1.000*	*1.000*	*33*
weighted avg	*1.000*	*1.000*	*1.000*	*33*

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	1	5	0.200
Double_cmere	2	5	0.400
Drum_Roll	1	5	0.200
Overall	**19**	**30**	**0.633**

Variant: CNN_BiLSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→BiLSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	105,510
CV mean ± std	0.9966 ± 0.0068
Hold-out test acc / loss	1.0000 · 0.0036
Batch-test acc	0.7667 (23/30)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	699	59	1.0000	1.0000	0.9831
2	699	59	0.9831	0.9831	0.9661
3	702	58	1.0000	1.0000	0.9828
4	702	58	1.0000	1.0000	0.9828
5	702	58	1.0000	1.0000	0.9483

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	1.000	1.000	1.000	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	*1.000*	*1.000*	*1.000*	*33*
weighted avg	*1.000*	*1.000*	*1.000*	*33*

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	4	5	0.800
Double_cmere	3	5	0.600
Drum_Roll	1	5	0.200
Overall	**23**	**30**	**0.767**